

ERP WEBAPP WORK FLOW.

1. Sales Management Module

1.1. Inquiry Received

- **Action:** Sales person enters new inquiry.
- **Required Fields:** Inquiry No (Auto), Customer Name (Ref: Customer Master), Inquiry Date (Date), Sales Person (User ID).
- **Items Array:** Item Name (Ref: Product), Quantity (Number), Target Price.
- **Options:** New, Processing, Quoted, Lost.

1.2. Stock Checking Logic

- **Action:** System calculation.
- **Fields:** Current Stock, Blocked Stock (Pending Orders), Net Available.
- **Logic:** If Net Available > Inquiry Qty, set Delivery Date = Today + 2 days. Else, Delivery Date = Today + Production Time.

1.3. Quotation

- **Action:** Send proposal.
- **Required Fields:** Quote ID, Inquiry Ref, Valid Until, Payment Terms.
- **Items:** Item, Qty, Rate, GST %, Total.

1.4. Receive Customer PO

- **Action:** Confirm order.
- **Required Fields:** PO Number (Customer's), PO Date, PO File (Upload), Confirmed Delivery Date.

1.5. Sale Order (SO)

- **Action:** Lock inventory.
- **Required Fields:** SO No, Customer PO Ref, Billing Address, Shipping Address.
- **Options:** Pending, Dispatched, Closed.

1.6. Sale Schedule & Dispatch Advice

- **Action:** Warehouse instruction.
- **Required Fields:** Dispatch ID, SO Ref, Transporter (Ref: Transport Master), Vehicle No, Driver Name.

1.7. Invoice

- **Action:** Bill generation.
- **Required Fields:** Invoice No, Date, Place of Supply, E-Way Bill No.
- **Financials:** Taxable Value, GST Amount, Round Off, Grand Total.

1.8. Automated Collection & Reminder System

- **Action:** System monitors due dates and sends auto-alerts.
- **Required Fields (Configuration):**
 - **Credit Period** (Days - Auto-fetched from Customer Master).
 - **Due Date** (Auto-calculated: Invoice Date + Credit Period).
 - **Reminder Settings** (Dropdown: Strict/Moderate/Lenient).
 - **Contact Person** (Ref: Customer Master - for WhatsApp/Email).
- **System Logic (Triggers):**
 - **Trigger 1: 7 Days Before Due Date** (Email & WhatsApp "Payment Upcoming").
 - **Trigger 2: 3 Days Before Due Date** (Email & WhatsApp "Payment Upcoming").
 - **Trigger 3: On Due Date** (Email & WhatsApp "Payment Due Today").
 - **Trigger 4: Overdue** (Every Days - Urgent WhatsApp/Email).
- **Log: Communication History** (Tracks "Sent Date", "Channel", "Status: Read/Delivered").

1.9. Voucher Receipt

- **Action:** Payment received.
- **Required Fields:** Receipt No, Date, Customer, Amount, Mode (Cheque/NEFT), Ref No.

2. Purchase Management Module

2.1. Material Indent

- **Action:** Internal Request.
- **Required Fields:** Indent No, Request Date, Department, Priority (High/Medium).
- **Items:** Item Name, Current Stock, Requested Qty.

2.2. Purchase Order (PO)

- **Action:** Order to Vendor.
- **Required Fields:** PO No, Vendor Name, PO Date, Valid Until.
- **Items:** Item, Qty, Rate, Expected Delivery Date.

2.3. Purchase Schedule

- **Action:** Track expected deliveries.
- **Required Fields:** PO Ref, Expected Date, Follow-up Status (On-Time/Delayed), Remarks.

2.4. GRN (Goods Receipt Note)

- **Action:** Gate Entry.
- **Required Fields:** GRN No, PO Ref, Vendor Challan No, Gate Entry Date, Vehicle No.

2.5. IQC (Incoming Quality Control)

- **Action:** Quality Check.
- **Required Fields:** GRN Ref, Item, Total Qty, Sample Size, Accepted Qty, Rejected Qty.

2.6. Material Receipt

- **Action:** Add to Stock.
- **Required Fields:** Receipt ID, GRN Ref, Storage Location (Rack/Bin), Batch No.

2.7. Purchase Billbook

- **Action:** Log Invoice.
- **Required Fields:** Bill ID, Vendor Invoice No, Invoice Date, Taxable Value, GST Amount, Total.

2.8. Voucher Payment

- **Action:** Pay Vendor.
- **Required Fields:** Voucher No, Payment Date, Vendor, Amount, Bank Account.

3. Production Management Module

3.1. Production Flow Details

- **Action:** Define BOM (Bill of Materials).
- **Required Fields:** Finished Good, Process Name (e.g., Cutting), Machine, Raw Material Input, Output Qty.

3.2. Production Routecard

- **Action:** Track batch progress.
- **Required Fields:** Route Card No, Batch No, Product, Plan Qty, Start Date, End Date.

3.3. Material Issue

- **Action:** Inventory to Floor.
- **Required Fields:** Issue ID, Route Card Ref, Item, Qty Requested, Qty Issued.

3.4. MTA (Material Transfer Acknowledgement)

- **Action:** Inter-department move.
- **Required Fields:** MTA No, From Dept, To Dept, Item, Qty, Received By.

3.5. Production Report

- **Action:** Daily Output Log.
- **Required Fields:** Date, Shift, Machine No, Operator, Production Qty, Rejection Qty.

3.6. Job Order (External)

- **Action:** Outsource work.
- **Required Fields:** Job Order No, Contractor, Item Sent, Process Required, Rate.

3.7. Challan Out

- **Action:** Send material out.
- **Required Fields:** Challan No, Job Order Ref, Item, Qty, Vehicle No.

3.8. External GRN & IQC

- **Action:** Receive processed goods.
- **Required Fields:** GRN No, Challan Ref, Received Qty, Passed Qty, Rejected Qty.

3.9. Job Work Billbook

- **Action:** Contractor Bill.
- **Required Fields:** Bill No, Job Order Ref, Labor Charges, GST.

3.10. Routecard Closure

- **Action:** Finish Batch.
- **Required Fields:** Route Card Ref, Final FG Qty, Scrap Generated, Closure Date.

4. Finance Management Module

4.1. Voucher Journal

- **Action:** Adjustment Entry.
- **Required Fields:** Journal No, Date, Debit Account, Credit Account, Amount, Narration.

4.2. Voucher Payment & Receipt

- **Action:** Bank/Cash transactions.
- **Required Fields:** Voucher Type (Payment/Receipt), Date, Party Name, Amount, Mode.

4.3. Voucher Contra

- **Action:** Cash Deposit/Withdrawal.
- **Required Fields:** Date, From Account (e.g., Cash), To Account (e.g., Bank), Amount.

4.4. Journal Voucher (GST)

- **Action:** Tax Adjustments.
- **Required Fields:** Date, GST Ledger (Input/Output), Adjustment Type (e.g., Reversal), Amount.

4.5. Bank Reconciliation

- **Action:** Match Bank.
- **Required Fields:** Bank Account, Statement Date, System Balance, Bank Balance, Unreconciled Amount.

4.6. Credit Card Statement

- **Action:** Expense Log.
- **Required Fields:** Card No, Statement Month, Transaction Date, Merchant, Amount, Expense Head.

5. HR Management Module

5.1. Employee Master

- **Action:** Add Staff.
- **Required Fields:** Emp Code, Name, Designation, Mobile, Joining Date, Basic Salary, Bank Details.

5.2. Salary Head Master

- **Action:** Define Components.
- **Required Fields:** Head Name (e.g., HRA, PF), Type (Earning/Deduction).

5.3. Employee Salary Structure

- **Action:** Assign Pay.
- **Required Fields:** Emp Name, Effective Date, Basic, HRA, DA, PF %.

5.4. Employee Salary Sheet

- **Action:** Monthly Payroll.
- **Required Fields:** Month, Year, Total Days, Present Days, Calculated Gross, Deductions, Net Pay.

5.5. Employee Advance Memo

- **Action:** Loan/Advance.
 - **Required Fields:** Emp Name, Date, Amount, Purpose, Recovery Month.
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6. Logistics Management Module

6.1. Transport Master

- **Required Fields:** Transporter Name, Owner Name, Mobile, GSTIN.

6.2. Transport Order

- **Action:** Book Truck.
- **Required Fields:** Order No, Transporter, Pickup Date, Destination, Vehicle Type.

6.3. Challan Out

- **Action:** Delivery Doc.
- **Required Fields:** Challan No, Transport Order Ref, Item List, Qty.

6.4. Freight Billbook

- **Action:** Transporter Invoice.
 - **Required Fields:** Bill No, Date, LR No, Freight Amount, GST, Total Payable.
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7. Quality Management Module

7.1. IQC (Incoming)

- **Required Fields:** GRN Ref, Item, Sample Qty, Visual Check (Pass/Fail), Dimension Check.

7.2. MTS (Material Transfer Slip)

- **Action:** Check during movement.
- **Required Fields:** MTA Ref, Item, Qty Checked, Status (OK/Damaged).

7.3. PQC (Process Quality Control)

- **Action:** In-process check.
- **Required Fields:** Route Card Ref, Stage Name, Operator, Observations.

7.4. PDI (Pre-Dispatch)

- **Action:** Final Audit.
- **Required Fields:** SO Ref, Box No, Packaging Condition, Label Accuracy.

7.5. QRD (Quality Rejection Decision)

- **Action:** Handle defects.
 - **Required Fields:** Rejection ID, Item, Qty, Action (Scrap/Return/Rework/Downgrade).
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8. Contractors Employee Management Module

8.1. Employee Master (Contractor Staff)

- **Action:** Register contract labor.
- **Required Fields:** Worker ID, Contractor Firm Name (Ref: Vendor), Worker Name, Skill Level (Skilled/Unskilled), Aadhar No.

8.2. Salary Head Master (Contractor)

- **Action:** Define rates.
- **Required Fields:** Role (e.g., Helper, Welder), Daily Rate, Overtime Rate.

8.3. Employee Salary Structure (Contractor)

- **Action:** Map worker to rate.
- **Required Fields:** Worker Name, Role, Applicable Daily Rate.

8.4. Employee Salary Sheet (Contractor)

- **Action:** Calculate Contractor Payout.
- **Required Fields:** Contractor Name, Month, Worker Name, Days Worked, Overtime Hours, Total Payable.
- **Logic:** $(\text{Days} * \text{Daily Rate}) + (\text{OT Hours} * \text{OT Rate})$.

8.5. Employee Advance Memo (Contractor)

- **Action:** Advances given to contractor for labor.
- **Required Fields:** Contractor Name, Date, Amount, Remarks.

8.6. Voucher Payment

- **Action:** Pay the Contractor.
 - **Required Fields:** Voucher No, Contractor Name, Salary Sheet Ref, Net Amount Paid, TDS Deducted.
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9. Maintenance Management Module

9.1. Tool Master

- **Required Fields:** Asset Code, Tool Name, Location, Maintenance Interval (Days).

9.2. Tool Maintenance Chart

- **Required Fields:** Tool Ref, Scheduled Date, Task List (Greasing, Cleaning).

9.3. Tool Calibration Report

- **Required Fields:** Tool Ref, Calibration Date, Standard Value, Actual Value, Result (Pass/Fail).

9.4. Tool Maintenance/Rectification Memo

- **Action:** Log repair.
 - **Required Fields:** Job ID, Tool Ref, Issue, Spares Used, Cost, Technician.
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10. Stores (Warehouse) Management Module

10.1. Warehouse Master

- **Required Fields:** Warehouse ID, Name, Address, Manager Name.

10.2. Warehouse Opening

- **Action:** Initial Stock Setup.
- **Required Fields:** Item Name, Opening Qty, Value, Date.

10.3. Dispatch SRV (Service Voucher)

- **Action:** Non-Sales Dispatch (e.g., Returnable Gate Pass).
- **Required Fields:** SRV No, Date, Party Name, Item, Qty, Return Expected Date (Yes/No).

10.4. Store/Warehouse Stock Transfer

- **Action:** Move between warehouses.
- **Required Fields:** Transfer ID, From Warehouse, To Warehouse, Item, Qty.

10.5. Warehouse Material Receipt

- **Action:** Receive transfer/return.
 - **Required Fields:** Receipt ID, Source Doc Ref, Item, Qty Received.
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11. Asset Management Module

11.1. Fix Asset Master

- **Required Fields:** Asset Tag, Name, Group (IT/Plant/Furniture), Purchase Date, Value.

11.2. Asset Addition Memo

- **Action:** Add new asset.
- **Required Fields:** Asset Ref, Invoice Ref, Installation Date, Depreciation Rate.

11.3. Asset Allocation Master

- **Action:** Assign to user.
- **Required Fields:** Asset Tag, Employee Name, Department, Date Assigned.

11.4. Asset Sale Memo

- **Action:** Dispose asset.
- **Required Fields:** Asset Tag, Sale Date, Sale Value, Book Value.

11.5. Asset Depreciation Voucher

- **Action:** Year-end calculation.
- **Required Fields:** Year, Asset Tag, Opening Balance, Depreciation Amount, Closing Balance.

12. Statutory Management Module

12.1. GST Taxation Master

- **Action:** Set Tax Rules.
- **Required Fields:** HSN Code, Description, IGST %, CGST %, SGST %.

12.2. GSTR-1 Upload (Sales)

- **Action:** Export for Filing.
- **Required Fields:** Month, Invoice No, Customer GSTIN, Taxable Value, Tax Amount, State.

12.3. GST2A Reconciliation (Purchase)

- **Action:** Compare Books vs Portal.
- **Required Fields:** Month, Vendor GSTIN, Total Input Tax Credit (ITC), Matched Amount, Mismatch Amount.

12.4. GST Deposit Challan

- **Action:** Tax Payment Record.
- **Required Fields:** Challan No, CPIN, Date, Bank, Tax Type (CGST/SGST), Amount.

12.5. TDS Trace & Details

- **Action:** Track Deduction.
- **Required Fields:** Section (e.g., 194C), Deductee Name, Payment Amount, TDS Rate, TDS Amount, Certificate No.

12.6. TCS Details

- **Action:** Tax Collected at Source.
- **Required Fields:** Customer Name, Sale Value, TCS Rate (e.g., 0.1%), TCS Amount.

12.7. GSTR1 & GSTR2 Register

- **Action:** Detailed Tax Ledger.
- **Required Fields:** Date Range, Transaction Type (B2B/B2C), Total Tax Liability.

12.8. Cheque Book Management

- **Action:** Manage Leaves.
- **Required Fields:** Bank Account, Start Leaf No, End Leaf No.
- **Status Tracker:** Leaf No, Status (Used/Cancelled/Blank), Issued To, Date.

12.9. Balance Sheet

- **Action:** Financial Statement.
- **Required Fields:** As On Date, Assets Total, Liabilities Total, Capital Account, Current Assets.

13. Production Simulation & Forecasting

We need a "**Production Simulation & Forecasting**" module in our ERP. This feature allows management to input a target production quantity (e.g., 20 Alto, 30 Swift) and immediately see the required raw materials, time estimates, and projected costs based on our current capacity.

1.Functional Flow

The user will input a **Master Production Schedule (MPS)**—a list of items and quantities. The system must perform a "Reverse Explosion" to calculate the following three outputs:

A. Material Requirements Planning (MRP)

- **Logic:** Fetch the *Bill of Materials (BOM)* for each target vehicle.
- **Calculation:** Target Quantity × BOM Quantity per Unit.
- **Output:** A consolidated list of all raw materials needed (e.g., Total Steel, Total Rubber) for the batch.
- **Inventory Check:** Highlight items where Required Qty > Current Stock.

B. Capacity Requirements Planning (CRP)

- **Logic:** Fetch the *Routing Table* (Standard Minute Value - SMV) for each vehicle.
- **Input Constraints:** The user defines the "Work Day" (e.g., 10 hours/day).
- **Calculation:**
 1. Total Man-Hours Required = $\sum(\text{Target Qty}) \times \text{Man-Hours per Unit}$.
 2. Days Required = Total Man-Hours / (Workforce Count × Shift Hours).

- **Output:** Estimated completion date and "Overload Alert" if the target cannot be met within the month.

C. Cost Estimation (Provisional Costing)

- **Labor Cost:** Total Man-Hours × Hourly Labor Rate.
- **Overhead (Electricity):** Total Machine Hours × Energy Rate (per kWh).
- **Material Cost:** Sum of Required Raw Material × Latest Purchase Price.

2. Database Requirements (Backend)

To support this, ensure the following relationships exist:

1. **Product Master:** Links to a **BOM Table** (ingredients) and a **Routing Table** (operations/time).
2. **Resource Master:** Stores Cost Per Hour for labor and Power Consumption (kWh) for machines.
3. **Shift Master:** Defines default working hours (e.g., 10 hours).

3. User Interface (UI) Requirements

- **Input Section:** A dynamic grid to add multiple products (Alto, Swift, Baleno) and set quantities.
- **Parameters:** Editable fields for "Shift Hours" and "Number of Workers" (defaulting to current settings).
- **Result Dashboard:**
 - **Summary Cards:** Total Cost, Total Days, Material Readiness (%).
 - **Detailed Grid:** Breakdown of raw materials needed vs. available.

4. Acceptance Criteria (Test Case)

- *Input:* 20 Alto, 30 Swift, 25 Baleno.
- *Constraint:* 10-hour shift, 50 workers.
- *Expected Result:* System must display exact raw material totals (e.g., X tons of steel) and precise labor cost (e.g., ₹X Lakhs) based on the defined BOM and Routing data.

Why this works:

- **"BOM" & "Routing":** These are the magic words. Developers know that "BOM" means the recipe (parts) and "Routing" means the steps (time).
- **"Reverse Explosion":** This tells them they need to start with the final car and calculate backwards to the nuts and bolts.

Project Scope & Confirmation Note

1. Workflow Verification We have outlined the ERP development workflow above based on your initial requirements.

- **Action Required:** Please review this workflow carefully. If you find any changes or missing steps, let us know immediately so we can adjust the project scope before we begin.

2. Automated Collection & Reminder System To improve cash flow and reduce manual follow-ups, we are introducing a **Smart Payment Reminder Module**:

- **Auto-Reminders:** You can set specific dates and frequencies for payment reminders based on your credit terms.
- **Multi-Channel Alerts:** The system will automatically send **WhatsApp and Email notifications** to your parties (customers) when their payment is due.
- **Customization:** You have full control to configure when and how often these reminders are sent, ensuring customers are notified exactly on time.

3. Integrated Communication Features

- **Direct Messaging:** Apart from reminders, you can send documents such as **Sales Invoices, Payment Vouchers, and Purchase Orders** directly to customers and vendors from the ERP dashboard.
- **Efficiency:** This seamless integration ensures faster delivery of important documents and better record-keeping.

4. Custom Reporting & Financials

- **Financial Statements:** The system will include automated **Profit & Loss** and **Balance Sheet** generation.
- **Custom Reports:** We will develop a dynamic **Report Section** tailored to your needs. Please provide us with the specific formats or types of reports you require (e.g., Daily Production, Sales Summaries), and we will integrate them.

5. Advanced Inventory Tools

- **Barcode System:** We will implement a feature to generate unique barcodes for all items. You can scan these barcodes to instantly identify products, making stock checking and dispatch faster and error-free.

6. Dashboard & Reports

- **Need to design every module graphically dashboard and 15-20 types of reports.**

7. Log in Authority

- **Super Admin can create number of user and give them power to access or view or both according to module or module and each page of module access.**

USE : React frontend with a Laravel backend